

The exact boundary-elliptic directions for a non- $\mathbb{C}$ -elliptic operator family

A partial result prompted by arXiv:2211.08167v1

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Result. For the operator $\mathbb{A}$ in Example 6.5 of Gmeineder–Raiță–Van Schaftingen, the exact set of boundary-elliptic unit directions is

$$\mathbb{S}^{n-1} \setminus (\text{span}\{e_1, e_2\} \cap \mathbb{S}^{n-1}).$$

The source records only the smaller sufficient set $\text{span}\{e_3, \dots, e_n\} \cap \mathbb{S}^{n-1}$. The exact characterization persists for every higher-order composition $\mathbb{B}\mathbb{A}$ constructed there with $\mathbb{B}$ complex-elliptic, and therefore gives an if-and-only-if classification of all halfspaces on which the source’s limiting trace and Sobolev estimates hold for this entire family.

1 Source question and operator

For a homogeneous constant-coefficient operator $\mathcal{D}$, write $\mathcal{D}[\zeta]$ for its complexified symbol. It is *boundary elliptic in direction* $\nu \in \mathbb{S}^{n-1}$ when

$$\ker_{\mathbb{C}} \mathcal{D}[\xi + i\nu] = \{0\} \quad \text{for every } \xi \in \mathbb{R}^n.$$

The source proposes future work on the set of admissible outward normals for general-domain traces, and closes with the following family [1].

For $n, N \geq 3$, the first-order operator $\mathbb{A}$ on $u = (u_1, \dots, u_N) : \mathbb{R}^n \rightarrow \mathbb{R}^N$ contains the equations

$$\begin{aligned} \partial_1 u_1 - \partial_2 u_2, & \quad \partial_1 u_2 + \partial_2 u_1, \\ \partial_j u_1, & \quad \partial_j u_2 & (3 \leq j \leq n), \\ \nabla u_q & & (3 \leq q \leq N). \end{aligned}$$

The source proves that $\mathbb{A}$ is elliptic and canceling but not $\mathbb{C}$ -elliptic, and notes boundary ellipticity for every direction in $\text{span}\{e_3, \dots, e_n\}$. It also composes $\mathbb{A}$ on the left with arbitrary complex-elliptic operators $\mathbb{B}$.

elliptic in a certain direction than to not be $\mathbb{C}$ -elliptic in general (see the next example), a modification of the argument sketched in Figure 2 directly yields that the existence of a trace operator $BV^{\mathbb{A}}(\Omega) \rightarrow T_k(\partial\Omega, V)$ forces the outward unit normals $\nu_{\partial\Omega}$ to belong to some set $K \subset \mathbb{S}^{n-1}$ depending on $\mathbb{A}$. While this technical point will be pursued in future work, we conclude the present paper by giving examples of operators that, in view of (6.1), fail to be $\mathbb{C}$ -elliptic, yet are boundary elliptic in certain directions and thus admit Sobolev estimates on the corresponding halfspaces:

Example 6.5. Let $n \geq 3$, $N \geq 3$, $V = \mathbb{R}^N$, $W = \mathbb{R}^{((N-1)n-1) \times 2}$ and consider the differential operator $\mathbb{A}$ acting on $u = (u_1, \dots, u_N): \mathbb{R}^n \rightarrow \mathbb{R}^N$

$$(6.9) \quad \mathbb{A}u = \left[\begin{array}{c|c} \partial_1 u_1 - \partial_2 u_2 & \partial_1 u_2 + \partial_2 u_1 \\ \partial_3 u_1 & \partial_3 u_2 \\ \vdots & \vdots \\ \partial_n u_1 & \partial_n u_2 \\ \hline \nabla u_3 & \mathbf{0} \\ \vdots & \vdots \\ \nabla u_N & \mathbf{0} \end{array} \right],$$

where $\mathbf{0}$ denotes the zero vector in $\mathbb{R}^n$. As established in [23, Counterexample 3.4], this operator serves as an example of an elliptic operator being canceling yet failing to be $\mathbb{C}$ -elliptic. However, $\mathbb{A}$ is boundary elliptic in every direction $\nu \in \text{span}\{e_3, \dots, e_n\}$. Based on this operator, boundary elliptic, non- $\mathbb{C}$ -elliptic operators of arbitrary order can be constructed: In fact, if $\mathbb{B}$ is a $(k-1)$ th order, $\mathbb{C}$ -elliptic differential operator on $\mathbb{R}^n$ from $W = \mathbb{R}^{((N-1)n-1) \times 2}$ to some finite dimensional real vector space, then $\mathbb{B}\mathbb{A}$ is of k th order, boundary elliptic in all directions $\nu \in \text{span}\{e_3, \dots, e_n\}$ but non- $\mathbb{C}$ -elliptic.

REFERENCES

- [1] ADAMS, R. A., 1975. *Sobolev spaces*. Pure and Applied Mathematics, Vol. 65. Academic Press, New York-London.
- [2] AGMON, S., DOUGLIS, A., and NIRENBERG, L., 1959. *Estimates near the boundary for solutions of elliptic partial differential equations satisfying general boundary conditions. I* Communica-

Figure 1: The future-work paragraph and Example 6.5 in arXiv:2211.08167v1, page 20. The source gives a sufficient family of directions; the theorem below identifies all of them.

2 Idea of the proof

All possible failure is concentrated in the first two input coordinates. The gradient rows kill $v_3, \dots, v_N$. If a complex covector has any nonzero coordinate beyond its first two, the remaining rows also kill v_1, v_2 . On the first coordinate plane, only the Cauchy–Riemann block remains, and its determinant factors into two complex linear forms. Its two characteristic lines can then be projected exactly onto their imaginary parts. This turns the algebraic characteristic variety into the desired geometric set of bad normals.

3 The characteristic cone

Theorem 1 (Exact characteristic variety). *For the operator $\mathbb{A}$ above,*

$$\{\zeta \in \mathbb{C}^n \setminus \{0\} : \ker_{\mathbb{C}} \mathbb{A}[\zeta] \neq \{0\}\} = \mathbb{C}^*(1, i, 0, \dots, 0) \cup \mathbb{C}^*(1, -i, 0, \dots, 0).$$

Consequently,

$$\mathbb{A} \text{ is boundary elliptic in direction } \nu \iff \nu \notin \text{span}\{e_1, e_2\}. \quad (1)$$

Proof. Fix $\zeta \neq 0$ and suppose $\mathbb{A}[\zeta]v = 0$, with $v = (v_1, \dots, v_N) \in \mathbb{C}^N$. For each $q \geq 3$, the gradient block gives $\zeta_j v_q = 0$ for every j . Since $\zeta \neq 0$, this forces $v_q = 0$. The remaining symbol equations are

$$\begin{pmatrix} \zeta_1 & -\zeta_2 \\ \zeta_2 & \zeta_1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0, \quad \zeta_j v_1 = \zeta_j v_2 = 0 \quad (3 \leq j \leq n).$$

If some $\zeta_j \neq 0$ for $j \geq 3$, then $v_1 = v_2 = 0$, so the kernel is trivial. Otherwise ζ lies in the first coordinate plane, where a nonzero kernel exists exactly when

$$\zeta_1^2 + \zeta_2^2 = (\zeta_1 + i\zeta_2)(\zeta_1 - i\zeta_2) = 0.$$

This proves the formula for the characteristic variety (and explicit nonzero kernel vectors verify both lines).

If ν has a nonzero coordinate ν_j , $j \geq 3$, every $\xi + i\nu$ lies outside these two lines, so $\mathbb{A}[\xi + i\nu]$ is injective for every real ξ . Conversely, take a unit vector $\nu = (a, b, 0, \dots, 0)$ and put $\xi = (b, -a, 0, \dots, 0)$. Then

$$\xi + i\nu = (b + ia, -a + ib, 0, \dots, 0) = (b + ia)(1, i, 0, \dots, 0),$$

which is a nonzero characteristic covector. Thus boundary ellipticity fails for every direction in the first coordinate plane, proving (1). $\square$

The exact set is open and dense in the sphere. In particular, it includes every mixed direction having both a component in $\text{span}\{e_1, e_2\}$ and a nonzero component orthogonal to that plane; these directions are not covered by the sufficient subspace recorded in the source.

4 Every order and the sharp estimates

Lemma 1 (Complex-elliptic left factors preserve the cone). *Let $\mathbb{B}$ be any homogeneous complex-elliptic operator whose domain is the target space of $\mathbb{A}$. Then for every $\zeta \in \mathbb{C}^n \setminus \{0\}$,*

$$\ker_{\mathbb{C}}(\mathbb{B}\mathbb{A})[\zeta] = \ker_{\mathbb{C}} \mathbb{A}[\zeta].$$

Proof. Constant coefficients give $(\mathbb{B}\mathbb{A})[\zeta] = \mathbb{B}[\zeta]\mathbb{A}[\zeta]$ (up to the harmless common symbol convention). Complex ellipticity says that $\mathbb{B}[\zeta]$ is injective. Hence $\mathbb{B}[\zeta]\mathbb{A}[\zeta]v = 0$ if and only if $\mathbb{A}[\zeta]v = 0$. $\square$

Corollary 1 (Exact admissible halfspaces). *Let $\mathcal{D} = \mathbb{A}$, or let $\mathcal{D} = \mathbb{B}\mathbb{A}$ be any order- k operator in the source's family. For a unit normal ν , each of the following holds if and only if $\nu \notin \text{span}\{e_1, e_2\}$:*

1. *the sharp trace estimate into the source's space $T_k(H_\nu, V)$;*

2. *the limiting halfspace Sobolev estimate*

$$\|D^{k-1}u\|_{L^{n/(n-1)}(H_\nu^+)} \leq C\|\mathcal{D}u\|_{L^1(H_\nu^+)} \quad (u \in C_c^\infty(\mathbb{R}^n, V)).$$

Proof. The lemma and Theorem 1 give the exact boundary-elliptic directions. The source's Theorems 1.2–1.3 identify boundary ellipticity with the appropriate first- or higher-order trace estimate. The operator $\mathbb{A}$ is real elliptic by Example 6.5, and $\mathbb{B}[\xi]$ is injective for every real $\xi \neq 0$, so $\mathbb{B}\mathbb{A}$ is real elliptic as well. Theorem 1.1 of the source therefore identifies the same normal set with the limiting Sobolev estimate. $\square$

5 Scope and novelty

This is a full characterization for the explicit arbitrary-order family, not for general operators or curved domains. In particular, it does not prove sufficiency of a pointwise normal condition after flattening a general boundary; that is part of the broader future-work program.

The current v1 PDF and source TeX state only the smaller sufficient set. A bounded search of the run's full-source indexes and exact web searches using the authors, the example, and the admissible-normal terminology found no later characterization or erratum. Novelty is provisional pending specialist review.

References

- [1] F. Gmeineder, B. Raită, and J. Van Schaftingen, *Boundary ellipticity and limiting L^1 -estimates on halfspaces*, arXiv:2211.08167v1 (2022).